

Trello Backend Capstone - Part 1

Overview

This project is a Trello-style backend application built using FastAPI, PostgreSQL, SQLAlchemy, and Alembic. It supports user registration, JWT authentication, board management, section management, ticket management, invitations, board membership, and role-based access control (RBAC).

Tech Stack

Python 3.12, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT Authentication, Uvicorn

Key Features

- User Registration and Login
- JWT Protected APIs
- Board Creation and Management
- Section CRUD Operations
- Ticket CRUD Operations
- Invitation-Based Collaboration
- Board Membership Management
- Role-Based Access Control (RBAC)

Setup

1. Clone the repository
2. Create and activate a virtual environment
3. Install dependencies
4. Configure environment variables
5. Run Alembic migrations
6. Start FastAPI using Uvicorn
7. Access Swagger UI at /docs

Part 1 Completion Summary

Implemented authentication, boards, sections, tickets, invitations, board membership, ticket ownership tracking, and RBAC. Verified functionality through Swagger UI testing.